

Offensive and Defensive Attention Reward in Reinforcement Learning for Football Game

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Abstract—In recent years, deep reinforcement learning (DRL) algorithms have made a lot of impressive progress in video games. The launch of Google Research Football (GRF) challenges these well-known DRL approaches in the territory of human-like sports games, which demands both the agent's individual competence and cooperation strategies with sparse rewards provided. Most of the current solutions have huge requirements in computing resources for self-play or high-quality expert data for imitation learning. In this work, we design an efficient algorithm to avoid these issues. Our method takes full advantage of the built-in AI robots which have a human-level cooperative intention but lack dribble and shoot abilities. The former intention is challenging for DRL to learn, while the latter abilities are feasible for DRL. In this paper to overcome the challenges, we first adopt the attention mechanism to model the offensive and defensive intention of the built-in AI. Secondly, we reshape the sparse reward of GRF by adding an auxiliary reward signal generated from the attention network outputs. Finally, the attention reward instructs the agent to produce more cooperative strategies in the DRL training. Experiments show that our attention reward generator can accelerate the GRF agent training process and our method achieves better performance than the on-policy DRL baseline in the single-agent 11 vs. 11 football scenario.

Index Terms—deep reinforcement learning, sparse reward, attention mechanism, agents coordination

I. INTRODUCTION

Deep reinforcement learning (DRL) methods have achieved numerous impressive results in various areas of game AI, such as Go [1], Atari [2], StarCraft II [3], [4], and other video games [5]. For a long time, sports game was an area of video games that had not been studied deeply by reinforcement learning (RL). In recent years, there has been a growing attraction to sports games among RL researchers. Football, honored as the king of sports, is a challenging game enjoyed by worldwide players. Hence the launch of Google Research Football (GRF) [6] facilitates the research on football game AI.

Because of the large state space, high stochasticity, and the cooperation demand, 11 vs. 11 GRF environment challenges both the micro adversarial competence and macro collaboration between multiple agents. Furthermore, the football game is a sparse reward [7] problem due to the relatively simple

rules and agents can obtain rewards merely by a few ways such as carrying the ball forward and shooting a goal, which makes the task even more arduous.

Considering of the large-scale and stochastic environment, some works have thought of ways to improve the performance of football player agents through self-play training methods [8], [9], with the drawback of long training time and high computational resource requirements. Another work improves the performance of multi-agent football players by using offline RL methods [10] and the drawback is one needs to primarily have a teacher model to provide high-quality replay data.

In a real-world football match, the offensive usually focuses on itself to dribble past an opponent and turns attention to the suitable teammate's position when a good chance of pass is to come. While the defensive concentrates on the opponent who is in the possession of the ball or to whom the ball is currently passed. This paper's idea is to model the offensive and defensive attention of football players during the instantaneous changes of state on the field, which can be treated as a significant estimation of the state values beneficial for RL training.

We think the intention of a football player to pass the ball or not at a certain time-step could be learned by neural networks. Particularly, attention mechanism, which has been proved to be effective in learning the latent relevance of sequential and group embedding vectors [11], is intuitively suitable for modeling this problem. Therefore, we build the offensive and defensive attention (ODA) model training with the data produced by the built-in AI and use it to describe the cooperative intention of the football game in our RL training. Then we design an offensive and defensive attention reward for the agent and transform the original sparse reward to be intensive. With the help of the widely used DRL algorithm named proximal policy optimization (PPO) [12], our method ODA-PPO can promote the active agent to cooperate with its teammates and achieve better performance than PPO baseline in 11 vs. 11 single-agent full football game.

II. RELATED WORKS

A. DRL Methods in Cooperative and Adversarial Scenarios

Training the active agent to play football like a human is one of the computational challenges for DRL methods due

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to the large-scale state space, sparse reward setting, and the difficulty in promoting agents to team play. Several impressive works in DRL have already been published to solve these large cooperative and adversarial scenarios as follows.

Some methods inspired by game theory [13]–[15] show their advantages in large state space and sparse reward scenarios. AlphaGo Zero [16] adopts a self-play paradigm and is able to train agents to master the skills of playing chess, shogi, and Go simultaneously without human data, which proves the huge potential of DRL combined with self-play training for the first time. The non-transitivity trait of some zero-sum games like football makes agent impossible to achieve high performance through a singular policy from self-play training [17]. Therefore, some researches [9], [18] concentrate on acquiring a diverse policy set which is originated from encouragements of behavioral diversity. And then the diverse policy set can exploit different opponents in sparse reward settings. Huge compute resource demand and time consumption are the pain point of this type of approaches.

From board games to large-scale team play games, state space drastically expands and the stochasticity steeply increases. Expert domain data gradually becomes a necessity for neural networks to initialize their parameters to benefit the subsequent self-play training procedure. AlphaStar [3] utilizes adequate expert human data through each part of a learning process which contributes a lot to the final result of achieving a Grandmaster level in StarCraft II. Imitation learning has been a significant scope in the intelligent decision and has attracted lots of attention [19]. Inverse reinforcement learning [20] reconstructs the reward function of reinforcement learning from the dataset to resolve the cumulative error of imitation learning. Generative adversarial imitation learning [21] simulates the expert policy with a generative adversarial network to reduce the reliance on the expert dataset. Recently, offline reinforcement learning [22] completely gets rid of the demand for expert data. Instead, the replay buffer dataset which is sampled through online DRL algorithms is enough for offline training to surpass the performance of the online baseline. TiKick [10] is an offline DRL method applied in GRF, which learns from Tencent WeKick [23] football single-agent RL-based model and transfers it to the multi-agent domain. However, the teacher model WeKick is the champion model of GRF Competition 2020 that coheres several advanced algorithmic and engineering technologies, so it is inaccessible to other researchers.

B. DRL Methods Combined with Attention Mechanism

Attention mechanism [24] has attracted great attention in the whole area of deep learning. With the joint efforts of many researchers, it has been successfully applied to natural language processing, image target detection, semantic understanding, etc [25]. Currently, some attempts are made to take advantage of the strong representative ability of the attention mechanism in DRL. Attention mechanism is capable of modeling the relevance between time sequence vectors or

group patch embedding, which are common in multi-agent reinforcement learning (MARL) [26].

In order to model the time sequence vectors in MARL, Ruan et al. [27] proposes a joint team policy graph generator realized by graph attention mechanism (GAT) [28] to learn an actor coordination graph for teamwork agents. Wen et al. [29] models the MARL as a sequence problem and designs a transformer [24] structure to find the solution. When time sequence vectors of each agent in a multi-agent system are well learned by the neural network, diverse cooperations between agents are hopeful to be seen soon.

In order to model the group patch members' embedding, Liu et al. [30] proposes a GAT aggregator to represent agents' states in the multi-agent system, aiming to let agents know which neighbors should be noticed more. Wu et al. [31] designs a two-level attention network. Firstly, it separates the whole group into friend group, foe group, and entity group. Subsequently, the first level attention network computes these different categories of observations. Ultimately, the second level attention network combines each kind of agents' attentive features. Attentive state representation has plenty of advantages such as the scalability in large-scale MARL, the permutation invariance between agents, and strong adaptation.

III. PRELIMINARIES

A. Markov Decision Process

A Markov decision process (MDP) consists of a tuple of five elements $\langle \mathbb{S}, \mathbb{A}, P, r, \gamma \rangle$, which can describe the process of an agent's interaction with an environment. Among the five elements, $\mathbb{S}$ means the set of state space and $\mathbb{A}$ denotes the action space for the agent. P defines the set of state transition probability distributions given by $P(s_{t+1}|s_t, a_t)$ over the current state s_t and action a_t . r represents the reward function $r : \mathbb{S} \times \mathbb{A} \times \mathbb{S} \rightarrow \mathbb{R}$ and $\gamma \in [0, 1]$ is the discount factor meaning the value of time. The goal for the agent is to maximize the discounted cumulative reward $R(s_t, a_t) = \mathbb{E}[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t, s_{t+1})]$. Mathematically, one solution for the agent is to find a policy function $\pi : \mathbb{S} \rightarrow \Delta(\mathbb{A})$, which means a probability distribution of $\mathbb{A}$. Therefore, the objective function for the agent to maximize can be expanded as:

$$\mathbb{E}_{s_{t+1} \sim P(\cdot | s_t, a_t)} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t, s_{t+1}) \mid a_t \sim \pi(\cdot | s_t), s_0 \right]. \quad (1)$$

Furthermore, the state value function over a specific state s and a stationary policy π can be described as the γ -discounted expected return:

$$V^\pi(s) \triangleq \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s, \pi \right] \quad (2)$$

where $\mathbb{E}[\cdot]$ describes the mathematical expectation of the strategy trajectories, r_t represents the reward computed by the reward function $r(s_t, a_t, s_{t+1})$ at time-step t and s_0 means the initial state to be assessed. Similarly, the action value function

describing the value of an action a over a specific state s through a stationary agent policy π can be defined as:

$$Q^\pi(s, a) \triangleq \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s, a_0 = a, \pi \right]. \quad (3)$$

Ultimately, the advantage function is defined as:

$$A^\pi(s, a) \triangleq Q^\pi(s, a) - V^\pi(s), \quad (4)$$

which normalizes the action value function and reduces the variance coming from the value function estimation. Reinforcement learning algorithms aim to optimize the agent policy and evaluate the value of the state exactly from different aspects to solve MDP problems.

B. Proximal Policy Optimization

Proximal policy optimization (PPO) is one of the widely used reinforcement learning methods that solve the policy optimization problem by simplifying the trust region policy optimization [32] approximately. The objective for the policy network of PPO in each update step is to minimize the loss function as follows:

$$\mathcal{L}(\theta) = \mathbb{E} \left[-\min \left(\rho_t(\theta), \text{clip}(\rho_t(\theta), 1 - \varepsilon, 1 + \varepsilon) \right) \tilde{A}_t \right] \quad (5)$$

where $\mathbb{E}[\cdot]$ describes the approximation of the policy distribution expectation through a batch of sampling trajectories generated by the current policy. ε is the clipping hyperparameter to control the update of the policy network, $\tilde{A}_t$ means the estimation of the advantage function $A^{\pi_\theta}(s_t, a_t)$ in which θ denotes parameters of the policy network, and $\rho_t(\theta)$ is the importance sampling weight defined by:

$$\rho_t(\theta) = \pi_\theta(a_t \mid s_t) / \pi_{\theta_{\text{old}}}(a_t \mid s_t) \quad (6)$$

where θ_{old} denotes the old policy network's parameters having participated in sampling. PPO algorithm takes advantage of monotonic improvement and robust performance in various environments guaranteed by trust region methods. Meanwhile, PPO saves calculation time substantially. Considering the dominance above, we choose the PPO algorithm as the baseline method in this paper.

C. Google Research Football

GRF is a suitable environment for researches in DRL because of high-dimensional inputs, high stochasticity, sparse reward settings, and so on. Moreover, GRF is a football game providing single-agent and multi-agent setting simultaneously. In single-agent mode, one can only control the active player closest to the ball and its teammates and opponents are all built-in AI in multiple levels differing from response speed.

The built-in AI robots provided by GRF enjoy the valuable perception of the current situation and play football through a combination of medium-level personal abilities and human-like team play by means of human football knowledge. Thus, the built-in AI acts as not only a good opponent for adversarial RL training but an ideal generator to establish a high-quality football intentional dataset.

GRF provides various scenarios such as 3 vs. 1, 5 vs. 5, and 11 vs. 11 full games for researchers to evaluate the performance of their algorithms. We choose the single-agent 11 vs. 11 scenario, which is the most universal and competitive scenario in football games, as the environment for our algorithm training and evaluation. The set of discrete actions with 19 elements is presented in TABLE I.

TABLE I
SET OF DISCRETE ACTIONS

Top Right	Bottom Top-Left	Left Top-Right
Bottom-Left	Bottom-Right	Shot
Short Pass	High Pass	Long Pass
Idle	Sliding	Dribble
Stop-Dribble	Sprint	Stop-Moving
Stop-Sprint	-	-

IV. METHODOLOGY

In this section, we will introduce the ODA-PPO algorithm in detail. The framework of the ODA-PPO given in Fig. 1 consists of the supervised learning procedure and the reinforcement learning procedure. The reinforcement learning procedure chooses the PPO algorithm to train the football game agent with the offensive and defensive attention reward generated from the outputs of the supervised training attention network. We will introduce each part of our work as follows.

A. Offensive and Defensive Attention Network

We want to design a structure of an attention network capable of learning the intention of football players through the inputs of some accessible state information such as the players' positions, velocities, directions, and so on. Moreover, when the offensive player keeps the ball, whether to pass the ball or not is also highly relative to the defensive threat given by opponent players' positions, velocities, directions, and so on. In contrast, to the defensive player, realizing who the ball is currently owned by or will be passed to are a couple of significant messages necessary to be judged immediately, which is similarly concerned with its teammates' current state. Considering these demands for offensive and defensive situation awareness, the attention network capable of aggregating the offensive and defensive information simultaneously is designed and the structure of the neural network is demonstrated in Fig. 2.

1) *Model Structure*: The offensive and defensive attention networks are comprised of two attention layers named opponents attention layer and teammates attention layer divided into left and right parts. For the sake of uniformity, hence the left team is treated as our team and the right team is the opponent. We first feed the original features into left players' encoder E^l and right players' encoder E^r implemented by fully connected neural network to get embedding vectors e_i^l and e_j^r

$$e_i^l = E^l(s_i^l) \quad (7)$$

$$e_j^r = E^r(s_j^r) \quad (8)$$

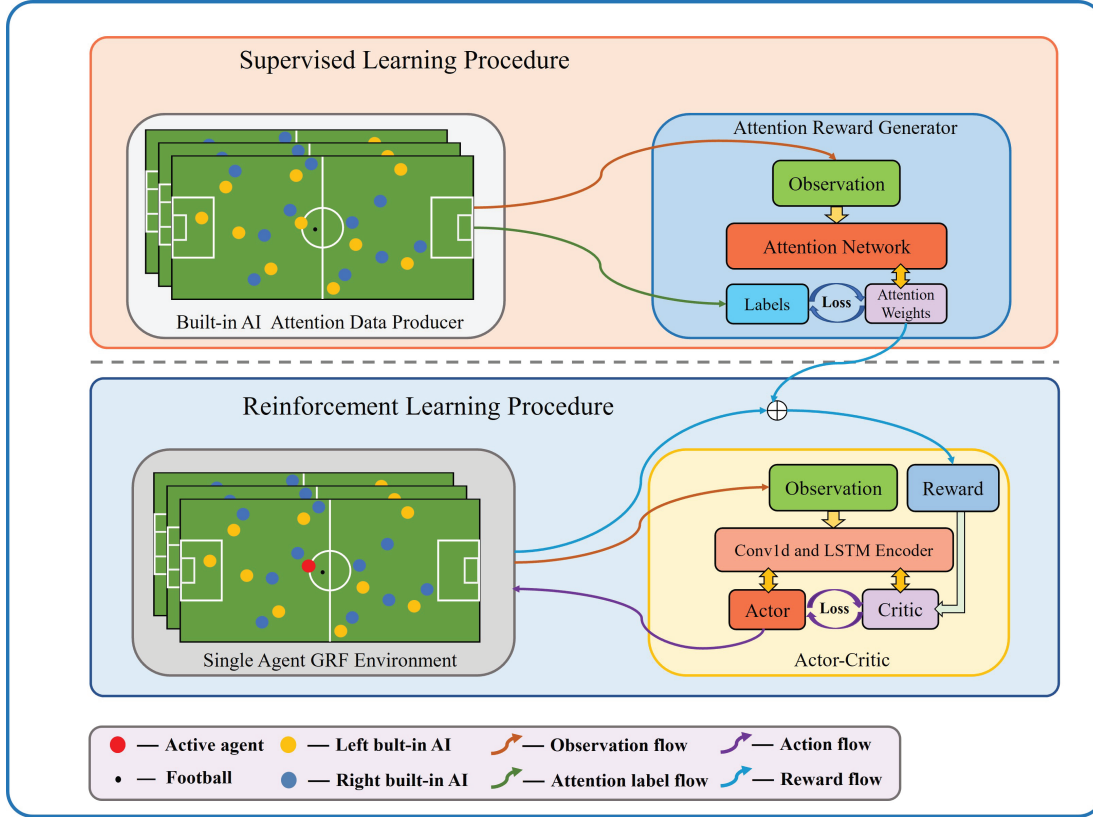


Fig. 1. The framework of football game agent reinforcement learning with offensive and defensive attention reward generator

where e_i^l means the state s_i^l embedding of the i -th player in 11 players of left team and e_j^r means the state s_j^r embedding of the j -th player in right team. Especially, the left team active player agent closest to the ball is represented as e_a^l where a denotes the active agent number between 1 and 11. When the ball is owned by the right team, it is inaccessible to know the owned-ball player number in a single-agent setting. Therefore, the ball state embedding named e_b^r is assigned to the right team to represent the state embedding of the owned-ball player in the right team.

Secondly, opponents attention layer computes attention scores separately for offensive network and defensive network inputs as follows:

$$e_{ij}^l = e_j^{rT} W_l^{kT} W_l^q e_i^l \quad (9)$$

$$e_{ji}^r = e_i^{lT} W_r^{kT} W_r^q e_j^r \quad (10)$$

where each left team player broadcasts a query transformed by W_l^q , while each right team member broadcasts a key transformed by W_l^k when the left team active player agent is on the offensive. In contrast, each right team player broadcasts a query transformed by W_r^q and each left team member broadcasts a key transformed by W_r^k in the defensive situation. Given the set of left team entities $\mathcal{L}$ and the set of right

team entities $\mathcal{R}$, the opponents attention values is computed as follows:

$$\alpha_{ij}^l = \frac{\exp(e_{ij}^l)}{\sum_{k \in \mathcal{R}} \exp(e_{ik}^l)} \quad (11)$$

$$\beta_{ji}^r = \frac{\exp(e_{ji}^r)}{\sum_{k \in \mathcal{L}} \exp(e_{jk}^r)} \quad (12)$$

The value of α_{ij}^l describes the defensive threat given to left player i by right player j and the meaning of β_{ji}^r is the defensive threat given to right player j by left player i . Opponents attention layer intends to integrate the opponents' information to the offensive and defensive from their own perspectives by concatenating their opponents' features after attention-weighted summation as follows:

$$g_i^l = \left[W_l^v e_i^l, \sum_{k \in \mathcal{R}} \alpha_{ik}^l W_l^v e_k^r \right] \quad (13)$$

$$h_j^r = \left[W_r^v e_j^r, \sum_{k \in \mathcal{L}} \beta_{jk}^r W_r^v e_k^l \right] \quad (14)$$

where W_l^v and W_r^v mean the value arrays for attention hidden vectors computation. The outputs of opponents attention network extract not only features of the player's own but the defensive threat features given by its opponents. These features are indispensable for the active agent to make a decision at the next time step.

Subsequently, the hidden attention features h_i^l and h_j^r are fed into the teammates attention network separately,

$$g_{ai}^{\text{off}} = g_i^l T W_{\text{off}}^k T W_{\text{off}}^q g_a^l \quad (15)$$

$$h_{bj}^{\text{def}} = h_j^r T W_{\text{def}}^k T W_{\text{def}}^q h_b^r \quad (16)$$

where W_{off}^k , W_{off}^q , W_{def}^k , and W_{def}^q are key-value matrixes and query-value matrixes for the second attention layer. The outputs of offensive and defensive attention networks are separately normalized as follows:

$$\alpha_{ai}^{\text{off}} = \frac{\exp(g_{ai}^{\text{off}})}{\sum_{k \in [1,11]} \exp(g_{ak}^{\text{off}})} \quad (17)$$

$$\beta_{bj}^{\text{def}} = \frac{\exp(h_{bj}^{\text{def}})}{\sum_{k \in [1,11]} \exp(h_{bk}^{\text{def}})} \quad (18)$$

where the attention scores computed by the teammates attention layer have distinct meanings. α_{ai}^{off} describes the offensive attention of the active agent to itself or its teammates in varying levels. When the active agent is in the defensive, it comes to β_{bj}^{def} to predict the probability of each opponent that the ball is flying to. The attention weights vectors $A_a^{\text{off}} = (\alpha_{ai}^{\text{off}})_{i \in [1,11]}$ and $B_b^{\text{def}} = (\beta_{bj}^{\text{def}})_{j \in [1,11]}$ constituted by attention scores are the outputs of the offensive and defensive attention network.

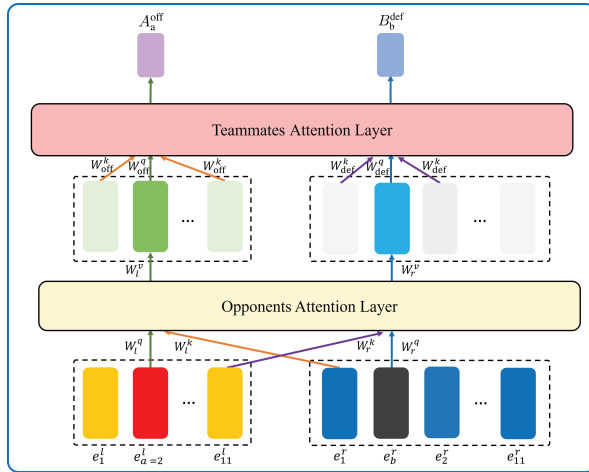


Fig. 2. The structure of offensive and defensive attention network

2) *Model Training*: After finishing the establishment of the network structure, it is imperative to provide the model with sufficient and informative data. The built-in AI designed by engineers through human knowledge in the field of football is considered to be an accessible resource to utilize, which has already possessed the fundamental potency in collaboration with teammates. The defects of the built-in AI are its technical skills in dribble, steal and shoot easy to learn through numerous repetitions methods such as reinforcement learning algorithms. GRF environment provides the built-in AI

masked action and the multi-agent setting, which is infeasible for imitation learning but convenient to get the high-quality intention data produced by the games between built-in AI agents.

We establish the attention data producer through built-in AI, which supplies the observation of each player's state as the input of the model and the record of the current game situation as the label in parallel. The records of games can be generally divided into the offensive situation and the defensive situation. In the offensive situation, the ball is controlled by the left team, which is opposite to the defensive. The offensive or defensive is further subdivided into dribble, shoot, pass, steal, catch the ball, and ball in the air. Only dribble and pass can depict the intention of an agent to its teammates through the one-hot encoding which codes itself or its teammate that the ball is passed to into vectors. Therefore, the model merely demands for dribble and pass inputs and labels to train the network to learn the intention of football players. Especially, considering the dribble data size far surpasses the pass data in competitions, we must control the proportional equilibrium between dribble and pass data to guarantee the network's well performance in two cases, which is set to 2:1 in the experiments. Summarily, the loss functions for offensive and defensive models training are depicted as:

$$\mathcal{L}^{\text{off}}(\varphi_1) = \mathbb{E}_{s_{\text{off}} \sim \mathcal{D}_{\text{off}}} \left[\left\| A_a^{\text{off}}(s_{\text{off}} | \varphi_1) - L_a^{\text{off}}(s_{\text{off}}) \right\| \right] \quad (19)$$

$$\mathcal{L}^{\text{def}}(\varphi_2) = \mathbb{E}_{s_{\text{def}} \sim \mathcal{D}_{\text{def}}} \left[\left\| B_b^{\text{def}}(s_{\text{def}} | \varphi_2) - L_b^{\text{def}}(s_{\text{def}}) \right\| \right] \quad (20)$$

where $\| \cdot \|$ represents the L_1 norm function. Meanwhile, $A_a^{\text{off}}(s_{\text{off}} | \varphi_1)$ denotes the offensive attention network output controlled by parameters φ_1 and a batch of offensive observation inputs s_{off} sampled from dataset $\mathcal{D}_{\text{off}}$. Similarly, $B_b^{\text{def}}(s_{\text{def}} | \varphi_2)$ means the defensive attention network outputs computed by parameters φ_2 and inputs s_{def} coming from the dataset $\mathcal{D}_{\text{def}}$. Moreover, $L_a^{\text{off}}(s_{\text{off}})$ and $L_b^{\text{def}}(s_{\text{def}})$ respectively represent the offensive and defensive batches of one-hot encoding labels generated from built-in AI. Both of them describe which player will keep the ball at the next time-step.

B. Policy Optimization with Attention Reward

1) *Proximal Policy Optimization*: Considering robustness and stability, we choose the PPO method as the baseline reinforcement learning algorithm. We use a form of PPO whose actor objective combines clipped advantages with policy entropy loss.

$$\mathcal{L}^{\text{actor}}(\theta) = \tilde{\mathbb{E}} \left[-\min \left(\rho_t(\theta), \text{clip}(\rho_t(\theta), 1 - \varepsilon, 1 + \varepsilon) \right) \tilde{A}_t - \zeta H(\pi_\theta(\cdot | s_t)) \right] \quad (21)$$

where $H(\pi_\theta(\cdot | s_t))$ that aims to promote exploration is defined as:

$$H(\pi_\theta(\cdot | s_t)) \triangleq \mathbb{E}_{s_t \sim \mathcal{D}, a \sim \pi_\theta} \left[-\ln \pi_\theta(\cdot | s_t) \right]. \quad (22)$$

To reduce the estimation variance of the advantage function and increase the sample efficiency, we use the generalized advantage estimator (GAE) [33] defined as follows:

$$\tilde{A}_t \triangleq \sum_{k=0}^N (\gamma \lambda)^k \delta_{t+k} \quad (23)$$

where $\lambda \in [0, 1]$ is a hyperparameter that controls the bias-variance trade-off in the process of N-steps advantage function estimation. δ_t represents the temporal difference (TD) error defined as:

$$\delta_t \triangleq r_t + \gamma V^\pi(s_{t+1}) - V^\pi(s_t) \quad (24)$$

where $V^\pi(\cdot)$ can be predicted by the value network of PPO and current reward r_t can be got from the environment. The objective for the critic network of PPO is to minimize the TD error which is defined as:

$$\mathcal{L}^{\text{critic}}(\omega) = \mathbb{E} \left[\left\| r_t + \gamma V_\omega(s_{t+1}) - V_\omega(s_t) \right\|^2 \right]. \quad (25)$$

Summarily, the overall loss function for the PPO algorithm to minimize can be integrated as follows:

$$\mathcal{L}(\theta, \omega) = \mathcal{L}^{\text{actor}}(\theta) + \mathcal{L}^{\text{critic}}(\omega). \quad (26)$$

The actor and critic parts of PPO are both constructed by fully connected neural network computed by parameters θ and ω . To model the representation of the team observations more completely, we choose one-dimension convolutional neural network (Conv1d) to aggregate the team state embedding and concatenate all the features. Subsequently, long short-term memory network (LSTM) is used to learn the time-sequential features whose outputs are fed into the actor and critic networks. Our PPO algorithm implementation shown in the reinforcement learning procedure of Fig. 1 consists of these components presented above.

2) *Attention Reward Generator*: The intention of playing football learned from built-in AI as mentioned above is supposed to be useful for reinforcement learning methods to promote the active agent to cooperate with its teammates. Through exploration, we find the offensive and defensive attention can be regarded as a reward generator for the active agent to learn better and faster. Detailly, the offensive reward that aims to promote team play is set to:

$$r^{\text{off}}(s_t, s_{t+1}) = \begin{cases} A_a^{\text{off}}(s_t)[i^*], & \text{if } A_a^{\text{off}}(s_t)[i^*] > 0.3 \\ 0, & \text{otherwise} \end{cases} \quad (27)$$

$$i^* = \text{Active}(s_{t+1}) \quad (28)$$

where a means the currently active player number and i^* means the active player number at the next time step after making a decision. Oppositely, the defensive reward is designed to predict the owned ball player in the right team as follows:

$$r^{\text{def}}(s_t, s_{t+1}) = \begin{cases} 0, & \text{if } d_{aj^*}(s_{t+1}) \leq 0.03 \\ -1, & \text{if } 0.03 < d_{aj^*}(s_{t+1}) \leq 0.06 \\ -2, & \text{if } 0.06 < d_{aj^*}(s_{t+1}) \leq 0.1 \\ -3, & \text{otherwise} \end{cases} \quad (29)$$

$$j^* = \text{argmax}(B_b^{\text{def}}(s_t)) \quad (30)$$

where j means the owned-ball player number in the right team predicted by the defensive attention network and $d_{aj^*} \in [0, 2]$ describes the distance between the active player and the j^* -th opponent player. The intention of the defensive reward encourages the agent to approach the ball and threaten the owned-ball opponent more. Summarily, the total reward including offensive and defensive situations is defined as:

$$r^{\text{tot}}(s_t, s_{t+1}) = \begin{cases} r^{\text{score}}(s_{t+1}) + 0.1r^{\text{off}}(s_t, s_{t+1}), & \text{if left team player owns the ball} \\ r^{\text{score}}(s_{t+1}) + 0.01r^{\text{def}}(s_t, s_{t+1}), & \text{otherwise} \end{cases} \quad (31)$$

$$r^{\text{score}}(s_{t+1}) = \begin{cases} +5, & \text{if score a goal} \\ -5, & \text{if lose a goal} \\ 0, & \text{otherwise} \end{cases} \quad (32)$$

where r^{off} has a higher weight in r^{tot} than r^{def} because the offensive pass is usually sparser in the training process than the defensive behaviors.

PPO with the help of ODA can accelerate the policy optimization process and reach a higher performance with more cooperative strategies than the baseline method. The results of the experiments will be shown in the next section.

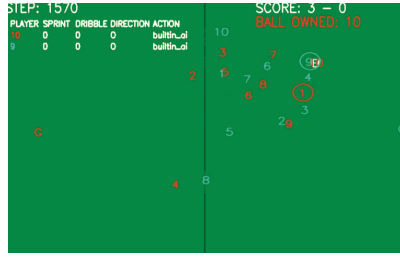
V. EXPERIMENTS

In this section, we will introduce the experiments' details to prove the effectiveness of our ODA-PPO method by comparing it with the PPO baseline algorithm in the 11 vs. 11 GRF single-agent scenario. The results of experiments will be divided into two parts which consist of attention network supervised learning and attention reward reinforcement learning.

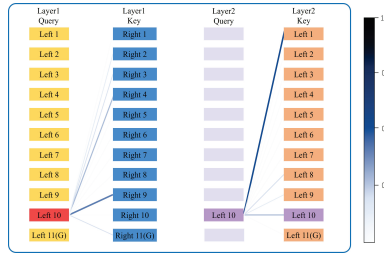
A. Attention Network Supervised Learning

Through over 10 million attentive data provided by built-in AI self-play, the offensive and defensive attention network converges and learns the intention of playing football successfully. From the demonstration of Fig. 3, we can see an example of the offensive and defensive attention learned by ODA model in a football competition, where ODA successfully predicts the most offensive teammate to cooperate with and the most defensive opponent that threatens the owning ball player. The outstanding performance of the attention network lays a solid foundation for the subsequent experiments in reinforcement learning with the attention reward.

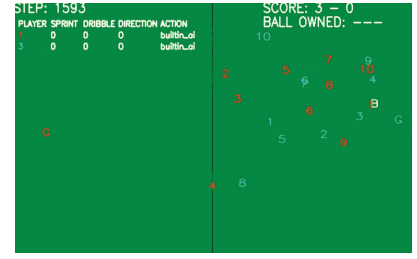
We can see the attention weight vectors entropy of the offensive and defensive attention network outputs in Fig. 4, which is conspicuous that the network has converged on the attention to specific players.



(a). Red 10 is dribbling the ball forward, focusing on the defensive threat given by Blue 9 and the offensive teammate Red 1 to pass the ball.



(b). This picture shows the attention scores of (a) computed by the offensive attention network in opponents layer and teammates layer.



(c). Red 10 passes the ball to Red 1 as it intends to do in (a) without facing any defensive threat from Blue opponents to shoot.

Fig. 3. Learning results demonstrations of the offensive and defensive attention network

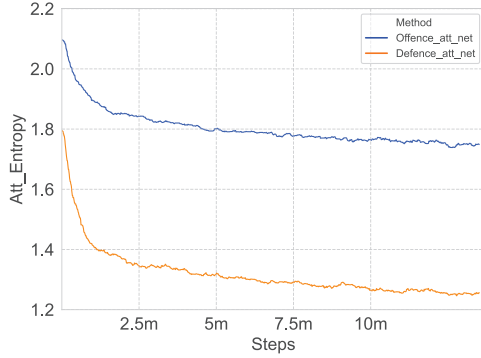


Fig. 4. Attention weights entropy of the offensive and defensive attention network

B. Attention Reward Reinforcement Learning

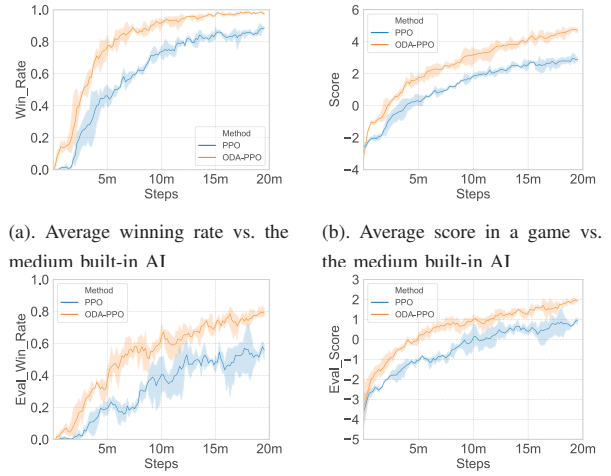
To accelerate the learning process, we choose to train the agents versus the middle difficulty built-in AI which relatively responds more slowly. While we evaluate the model performance with the strongest competitive built-in AI. The hyperparameters settings for RL training are listed in TABLE II.

TABLE II
HYPERPARAMETERS FOR RL TRAINING

Categories	Numerics
Time window length	30
l_r	0.0001
γ	0.993
λ	0.96
ζ	0.0001
ϵ	0.1
Batch size	32

After exploiting nearly 20 million training data to train the agent, ODA-PPO steadily achieves a higher winning rate and more scores both in the training process and in the evaluation as shown in Fig. 5 and Fig. 6.

From Fig. 5(a)-(b), we can see the ultimate winning rate against the medium AI of ODA-PPO surpasses 10% more



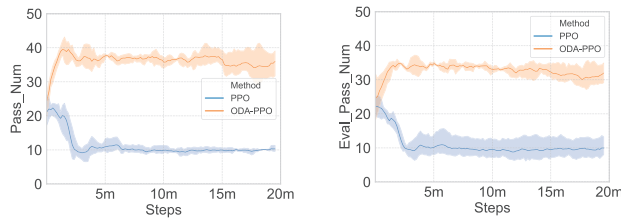
(a). Average winning rate vs. the medium built-in AI (b). Average score in a game vs. the medium built-in AI

(c). Evaluation of average winning rate vs. the competitive built-in AI (d). Evaluation of average score vs. the competitive built-in AI

Fig. 5. Results of winning rates and scores vs. the medium difficulty AI and the competitive AI

than that of the PPO baseline along with two more goals each game. When models are evaluated against the competitive AI in Fig. 5(c)-(d), ODA-PPO increases its policy more stably and defeats the hardest opponent with a winning rate of more than 80% lastly.

The most obvious feature of ODA-PPO model is the collaborative strategy reflected by the average successful pass numbers in a game. From Fig. 6(a) and Fig. 6(b), we clearly discover that the strategies learned from the PPO baseline are monotonous and lacking in collaboration. It seems that due to the defect of built-in AI opponents in velocities and steal, the defensive deterrence faced by the reinforcement learning agent is relatively weak, which causes the policy converges on greedy styles like dribbling the ball from back to front and shooting. While the policy learned by ODA-PPO is more reasonable and entirely different from the baseline model, which on average has more than twenty nice-pass numbers in a game than the PPO baseline.



(a). Average pass numbers vs. the medium built-in AI (b). Evaluation of average pass numbers vs. the competitive built-in AI

Fig. 6. Results of successful pass numbers vs. the medium difficulty AI and the competitive AI

VI. CONCLUSION

We provide a novel way to combine expert knowledge with reinforcement learning in the domain of team-play sports games. We use the intention data from the built-in AI to model the attention of a football player in a football sports game. By learning from expert knowledge, ODA model reconstructs the attentive reward that is able to guide the agent when to dribble and when to share the ball. As the results of the experiments, cooperation between the agent and its teammates can be promoted by the attentive reward and the performance of our ODA-PPO model can gradually surpass the built-in AI and the PPO baseline through the reinforcement learning process.

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